Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/23/2006
PI Name:	Fisher, John W M.S.		
Project Title:	Development of Metal Impregnated Single Walled Carbon Nanotubes (SWCNT) for Elimination of Toxic Contaminants in Advanced Life Support Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Moffet Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/30/2003	**End Date:**	01/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Li, Jing (Eloret at NASA Ames) Wignarajah, Kanapathipillai (EASI at NASA Ames) Cinke, Martin (Eloret at NASA Ames) Partridge, Harry (NASA Ames)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A number of waste management technologies are being developed that stabilize waste or recover resources for long duration manned missions. The success of many waste processing resource recovery technologies thus depends on the ability of gas clean-up systems to efficiently remove contaminants produced by the waste management systems. There is reason to believe that a new approach to gas clean-up using carbon nanotubes may have significant advantages in mass, power, and volume over traditional approaches. Carbon nanotubes possess selective adsorptive and catalytic properties as well as high surface areas that suggest that significant improvements can be made over currently available methods for gaseous contaminant cleanup. Single walled carbon nanotubes (SWCNTs) are currently being studied by a team at NASA Ames Research Center for their effectiveness as catalysts and catalyst support materials for trace contaminant control. The research on this project at NASA Ames focuses on: purification of single walled carbon nanotubes (SWCNT); catalyst impregnation of SWCNTs; characterization and application of SWCNTs as catalysts and/or catalyst supports for toxic gas conversion; and development of a prototype reactor. The single walled carbon nanotubes synthesized by catalytic methods contain metallic impurities such as Fe, Ni, and Co in addition to various forms of carbon such as graphite, amorphous carbon, and nanoparticles. Purification involves removal of these impurities, which results in a high surface area, molecularly well-defined SWCNT. Several different methods for purification to about 0.5% residual metal are underway. For catalyst applications, impregnation of the SWCNTs with metals such as Rh, Pd, Pt, Ag, Ru, Nb is being conducted. Analytical information on nanostructure and composition is being accumulated for correlation with results from adsorption and conversion efficiency experiments. A NASA Ames research team with combined expertise in nanotechnology research and development of advanced life support systems is investigating the potential applications of single walled carbon nanotubes for trace contamination control for future manned space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential benefits to Earth include production of improved contaminant adsorption and removal systems for buildings, cars, airplanes, and ships, and industrial processes.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)